



Name: _____

Date: _____

Unit 3: Thermochemistry Unit TestKnowledge (20 marks) - Multiple Choice on D2LThinking (18 marks)

1. The following data was collected for four trials of the following experiment: (7 marks total)

Experiment	Initial [A] mol/L	Initial [B] mol/L	Initial [C] mol/L	Rate mol/L*s
1	0.10	0.10	0.10	8.0×10^{-4}
2	0.20	0.10	0.10	1.6×10^{-3}
3	0.10	0.20	0.10	1.6×10^{-3}
4	0.10	0.10	0.20	3.2×10^{-3}

a) Use this information to determine the order of the reactants. Show your work! (3 marks)[A]^x use trial 1 + 2[B]^y use trial 1 + 3[C]^z use trial 1 + 4

$$\left(\frac{0.20}{0.10}\right)^x = \left(\frac{1.6 \times 10^{-3}}{8.0 \times 10^{-4}}\right)$$

$$\left(\frac{0.20}{0.10}\right)^y = \left(\frac{1.6 \times 10^{-3}}{8.0 \times 10^{-4}}\right)$$

$$\left(\frac{0.20}{0.10}\right)^z = \left(\frac{3.2 \times 10^{-3}}{8.0 \times 10^{-4}}\right)$$

$$2^x = 2 \quad \checkmark$$

$$x = 1$$

$$2^y = 2 \quad \checkmark$$

$$y = 1$$

$$2^z = 4 \quad \checkmark$$

$$z = 2$$

[A] is order 1

[C] is order 2

[B] is order 2

b) Write the rate law equation. (1 mark)

$$\text{rate} = k[A][B][C]^2 \quad \checkmark$$

c) What is the value of the rate constant, k? (2 marks)

$$\text{rate} = k[A][B][C]^2$$

$$k = \frac{\text{rate}}{[A][B][C]^2}$$

$$= \frac{8.0 \times 10^{-4} \text{ mol/L}\cdot\text{s}}{(0.10 \text{ mol/L})(0.10 \text{ mol/L})(0.10 \text{ mol/L})^2}$$

$$= \frac{8.0 \times 10^{-4} \text{ mol/L}\cdot\text{s}}{(0.10 \text{ mol/L})(0.10 \text{ mol/L})(0.10 \text{ mol/L})^2}$$

$$= 8 \text{ L}^2 / \text{mol}^2 \cdot \text{s}$$

$$k = \frac{8.0 \text{ L}^2}{\text{mol}^2 \cdot \text{s}}$$

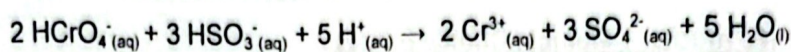
d) What is the overall order for the reaction? (1 mark)

$$1 + 1 + 2 = 4 \quad \checkmark$$

$$\frac{\text{L}^3}{\text{mol}^3 \cdot \text{s}}$$

order - 1 = exponent of unit

2. For the reaction below, the rate of consumption of the HSO_3^- ions was $2.4 \times 10^{-2} \text{ mol/L}\cdot\text{s}$. What are the rates of formation of the sulfate ion and the water in this experiment? (2 marks)



$$\text{SO}_4^{2-}: \frac{[\text{SO}_4^{2-}]}{\Delta t} = \frac{3 \text{ mol HSO}_3^-}{2.4 \times 10^{-2} \text{ mol/L}\cdot\text{s}} \times \frac{3 \text{ mol SO}_4^{2-}}{3 \text{ mol HSO}_3^-} = 2.4 \times 10^{-2} \text{ mol/L}\cdot\text{s}$$

$$\text{H}_2\text{O}: \frac{[\text{H}_2\text{O}]}{\Delta t} = \frac{3 \text{ mol HSO}_3^-}{2.4 \times 10^{-2} \text{ mol/L}\cdot\text{s}} \times \frac{5 \text{ mol H}_2\text{O}}{3 \text{ mol HSO}_3^-} = 0.04 \text{ mol/L}\cdot\text{s}$$

so the rate of formation
for SO_4^{2-} is $2.4 \times 10^{-2} \text{ mol/L}\cdot\text{s}$ and H_2O is $4.0 \times 10^{-2} \text{ mol/L}\cdot\text{s}$

3. When a 20.0 g sample received 47.0 J of energy, its temperature changed from 25.0°C to 35.0°C . Was this substance made from gold, lead or silver? Look up the specific heat capacities and make sure to include a rearranged formula in your answer. (3 marks)

$$q = mc\Delta t$$

$$c = \frac{q}{m\Delta t}$$

$$\text{gold } c: 0.129 \text{ J/g}\cdot^\circ\text{C}$$

$$\text{lead } c: 0.159 \text{ J/g}\cdot^\circ\text{C}$$

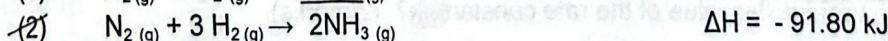
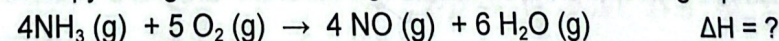
$$\text{silver } c: 0.237 \text{ J/g}\cdot^\circ\text{C}$$

$$c = \frac{47.0 \text{ J}}{(20.0 \text{ g})(35.0^\circ\text{C} - 25.0^\circ\text{C})}$$

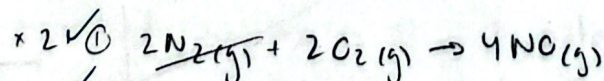
$$= 0.235 \text{ J/g}\cdot^\circ\text{C}$$

so the substance is silver

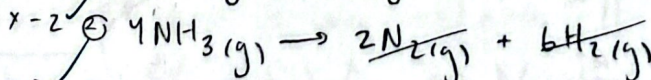
4. Calculate the enthalpy change for the following reaction from the following equations: (6 marks)



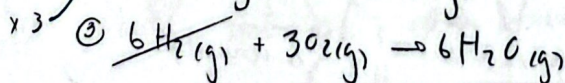
show
full work
like this
on exam



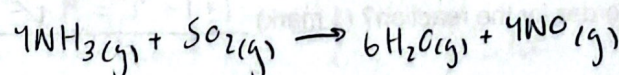
$$\Delta H = -361 \text{ kJ}$$



$$\Delta H = 183.6 \text{ kJ}$$



$$\Delta H = -1450.8 \text{ kJ}$$



$$\Delta H = -1628.2 \text{ kJ}$$

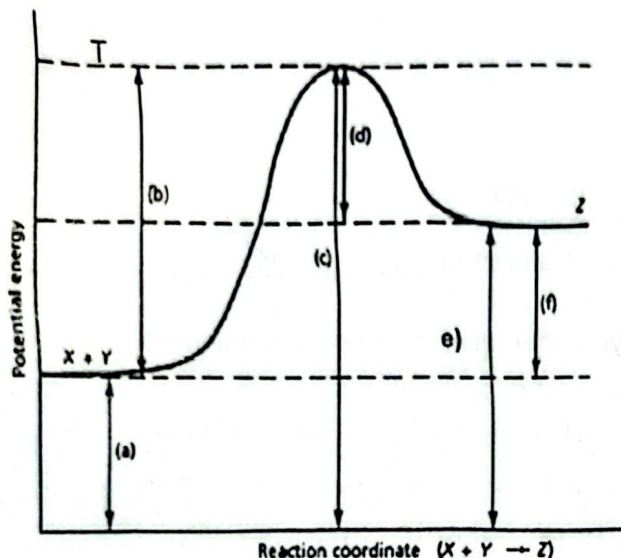
$$-1628$$

$$\uparrow$$

Communication (10 marks)

1. Examine the following Potential Energy Diagram and answer the following questions.

- a) Is this reaction exothermic or endothermic? (1 mark) endothermic ✓
 b) Indicate the letter on the diagram that best represents the following statements. (5 marks)



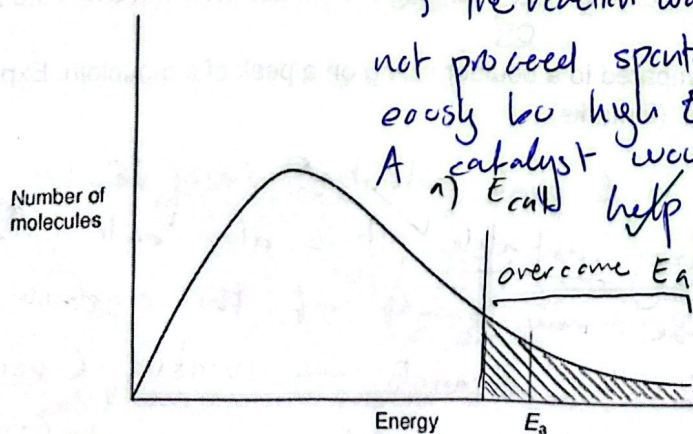
e) ✓	Which of the letters in the diagram represents the potential energy of the products ?
c) ✓	Which letter indicates the potential energy of the activated complex ?
b) ✓	Which letter indicates the activation energy ?
a) ✓	Which letter indicates the potential energy of the reactants ?
f) ✓	Which letter indicates the heat of reaction ?

c) Would this reaction be spontaneous and proceed without assistance? **Provide an explanation.** (2 marks)

It can proceed. → would not proceed, too high E_a
 Yes, this reaction could be spontaneous and proceed without assistance. The reactants would have to have enough kinetic energy for collisions and collision geometry to overcome activation energy. However, with the assistance of a catalyst,

2. The diagram below shows the Maxwell-Boltzmann distribution for a sample of a gas at a fixed reaction temperature. E_a is the activation energy for the decomposition of this gas. (2 marks) rate would ↑.

- a) Label the activation energy for the catalyzed reaction using the symbol E_{cat} and shade in the corresponding areas.
 b) With reference to the Maxwell-Boltzmann distribution, explain how using a catalyst leads to an increase in the reaction rate.



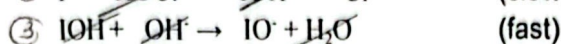
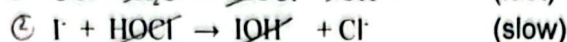
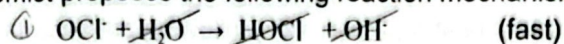
the reaction would not proceed spontaneously because of high E_a . A catalyst would help it proceed.

b) catalysts provide an alternative path to reaction with lower activation energy required. So, due to ↓ E_a , a greater portion of molecules have sufficient energy to overcome the E_a barrier, leading to more collisions, resulting in an increase of reaction rate.

+ more molecules with correct geometry

Application (10 marks)

1. A chemist proposes the following reaction mechanism:



a) Write the overall reaction equation. (1 mark)



b) List any reaction intermediate(s) and/or catalyst(s). (2 marks)

Intermediate(s): HOCl, OH⁻, IOH⁻Catalyst(s): H₂O

c) Given a proposed reaction mechanism, can you differentiate between a reaction intermediate and a catalyst? Explain. (2 marks)

you can differentiate between intermediate and catalyst by the following. Intermediates are formed in the elementary steps, and used up as a reactant in later steps. Catalysts are not formed, but appear as a reactant then product.

d) The rate law equation was determined to be $\text{rate} = k[\text{I}^-][\text{HOCl}]^2$. Is the proposed mechanism plausible? Support your answer. (1 mark)

no, this is not plausible because the rate determining step (slowest) is $\text{I}^- + \text{HOCl}$, where they both are @ 1 mol (coefficient), so they would have corresponding order 1, not $\text{rate} = k[\text{I}^-][\text{HOCl}]^2$

e) A new catalyst is discovered that decreases the activation energy for step 3. How will this affect the overall rate of the reaction? Give a reason for your answer. (2 marks)

this will not affect the overall rate of reaction, as the overall rate of reaction is determined by the slowest step (step 2) because of its high E_a , so lowering E_a for step 3 won't do anything. To increase overall reaction rate, E_a must ↓ for step 2

2 a) Use collision theory and a diagram to explain how changing the surface area can affect the rate of a reaction. (2 marks)

OR

b) An activated complex has been compared to a boulder sitting on a peak of a mountain. Explain why this is an appropriate comparison. (2 marks)

b) an activated complex is @ peak potential energy (@ highest point of the hill) and is very unstable. It is also called the transition state because new bonds of the products are forming and bonds of the reactants are breaking (neither fully product or reactant), so like a boulder @ the peak (transition from  are side to another, but not on either side fully yet)